JOSE MARTIN IMPERIAL
BSIT - 4D
WEEK 2 – PORTFOLIO PROJECT

PART 1: COMPANY PROFILE

Company Name:

NexaCore Solutions

Nature of Business:

NexaCore Solutions is a small software development company that builds websites, mobile apps, and IT solutions for clients.

Company Vision:

To become a leading regional provider of innovative, reliable software solutions that empower businesses to grow digitally.

Company Mission:

To deliver high-quality, secure, and scalable software products through skilled talent, sound infrastructure, and strong client partnerships.

Office Location:

4th Floor, Innovate Business Center, Brgy. San Isidro, Sta. Rosa City, Laguna, Philippines

Organizational Structure

Department	Employees
Information Technology	5
Human Resources	4
Finance	5
Sales	6
TOTAL	20

Employee Distribution

- IT Department: 5 employees
- HR Department: 4 employees

- Finance Department: 5 employees
- Sales Department: 6 employees

The IT Department will manage the company's computers, network, servers, software, and security.

PART 2: ENTERPRISE HARDWARE INVENTORY

Asset ID	Hardware	Quantity	Department	Purpose
HW-001	Desktop Computer	12	IT, Finance, HR	Fixed workstations for daily office/dev tasks
HW-002	Laptop	8	Sales, Management	Mobility for client visits and presentations
HW-003	Server (Tower)	1	IT	Hosts file sharing, domain, backups
HW-004	Router	1	IT	Manages internal/external network traffic
HW-005	Switch (24-port)	2	IT	Connects all wired devices per floor
HW-006	Printer (Network)	2	HR, Finance	Shared document printing
HW-007	UPS (1500VA)	3	Server room, key desks	Power protection during outages
HW-008	Wireless Access Point	2	Whole office	Wi-Fi coverage for laptops/mobile devices
HW-009	NAS Storage (4TB)	1	IT	Centralized file storage and redundancy
HW-010	External Backup Drive	2	IT	Offline backup rotation
HW-011	Monitor	12	IT, Finance, HR	Paired with desktops

Justification: Quantities match the 20-employee headcount (12 desk-based staff get desktops, 8 mobile/managerial staff get laptops). Two switches and two access points ensure coverage and redundancy on a single floor. UPS units protect the server and critical desks from frequent brownouts.

Computer Specifications:

Desktop Computers

- Intel Core i5 / AMD Ryzen 5

- 16 GB RAM
- 512 GB SSD
- Windows 11 Pro
- Gigabit Ethernet

Laptops

- Intel Core i5 / Ryzen 5
- 16 GB RAM
- 512 GB SSD
- Windows 11 Pro
- Wi-Fi 6

Server

- Intel Xeon or similar server-grade CPU
- 32–64 GB RAM
- 2–4 TB storage (RAID configuration)
- Ubuntu Server
- Dual network ports

These specifications provide enough headroom for normal office work, programming, file sharing, and basic IT tasks, while allowing room to grow as the company scales.

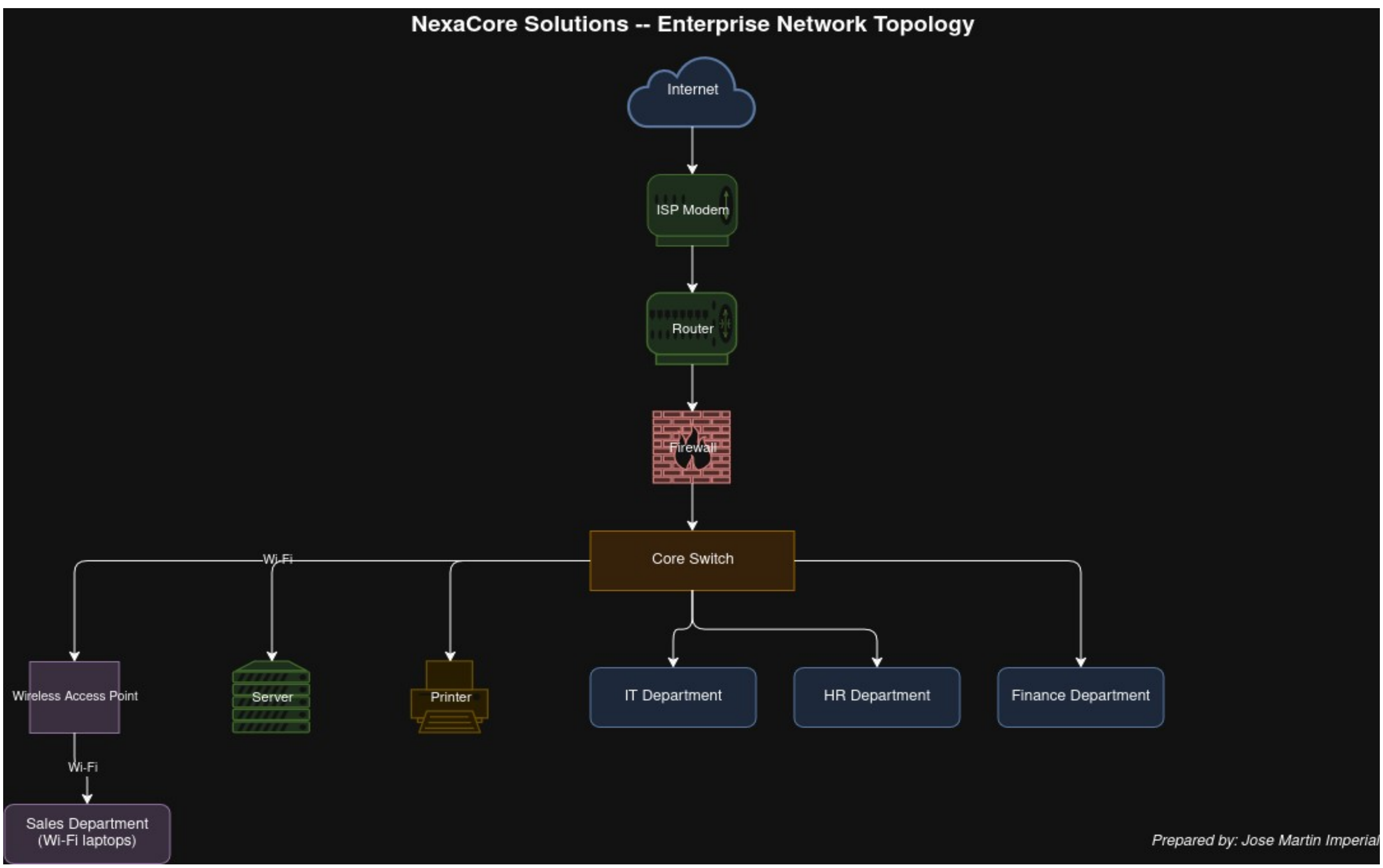
PART 3: ENTERPRISE SOFTWARE INVENTORY

Software	Version	License	Purpose
Windows 11 Pro	23H2	Volume license (20 seats)	Primary OS for desktops/laptops
Ubuntu Server	22.04 LTS	Free / Open-source	Hosts internal apps, file server, dev/test environment
Microsoft Office	365 Business Standard	Subscription (20 seats)	Word processing, spreadsheets, email, collaboration
VS Code	Latest	Free	Primary IDE for the development team
Git	Latest	Free	Version control for source code
GitHub Desktop	Latest	Free	GUI for repository management
VirtualBox	Latest	Free	Local testing/staging environments
Google Chrome	Latest	Free	Standard web browser
Microsoft Defender	Built-in	Included with Windows	Endpoint antivirus/anti-malware
AnyDesk	Latest	Free (business tier optional)	Remote support/helpdesk access
7-Zip	Latest	Free	File compression/archiving

PART 4: ENTERPRISE NETWORK INVENTORY

Asset ID	Network Equipment	Quantity	Purpose
NET-001	ISP Modem	1	Receive internet connection
NET-002	Router	1	Manage network traffic
NET-003	Firewall	1	Protect the company network
NET-004	Managed Switch	2	Connect computers and devices
NET-005	Wireless Access Point	2	Provide Wi-Fi
NET-006	Patch Panel	1	Organize network cables
NET-007	CAT6 Cable	1 lot	Connect network devices
NET-008	RJ45 Connectors	1 lot	Connect network cables

PART 5: ENTERPRISE NETWORK DIAGRAM



PART 6: SYSTEM ADMINISTRATION ROLES

1. Helpdesk Technician

Responsibilities: First-line troubleshooting, ticket handling, hardware/software setup, password resets.

Skills: Communication, basic OS/network troubleshooting, patience under pressure.

Common Tools: Ticketing systems (Zendesk, Freshservice), AnyDesk/TeamViewer, Active Directory Users & Computers.

Certifications: CompTIA A+, ITIL Foundation.

2. Network Administrator

Responsibilities: Manage routers/switches/firewalls, monitor bandwidth, ensure uptime and security of the network.

Skills: Subnetting/VLANs, routing protocols, firewall configuration, network monitoring.

Common Tools: Wireshark, PRTG/Nagios, Cisco Packet Tracer, pfSense.

Certifications: CompTIA Network+, CCNA.

3. Linux System Administrator

Responsibilities: Maintain Linux servers, manage users/permissions, automate tasks, apply patches, ensure server uptime.

Skills: Bash scripting, package management, systemd, log analysis.

Common Tools: SSH, cron, Ansible, systemctl, Vim/Nano.

Certifications: CompTIA Linux+, RHCSA.

4. Cloud Administrator

Responsibilities: Manage cloud infrastructure (VMs, storage, networking), cost optimization, cloud security, backups.

Skills: IaaS/PaaS concepts, scripting/automation, cloud networking, security groups/IAM.

Common Tools: AWS/Azure/GCP consoles, Terraform, CloudWatch/Azure Monitor.

Certifications: AWS Certified Solutions Architect, Microsoft AZ-104.

How These Roles Work Together

In a small organization like NexaCore Solutions, these four roles form a support chain — the Helpdesk Technician handles day-to-day user issues and escalates unresolved or infrastructure-level problems to the Network Administrator, who keeps connectivity and security running, and the Linux System Administrator, who maintains internal servers. As the company grows and adopts cloud services, the Cloud Administrator would take over hosting, scalability, and offsite backup responsibilities, coordinating with the Network Administrator on secure connectivity between on-premise and cloud resources. Together they ensure the company's IT operations remain available, secure, and scalable as headcount grows.

PART 7: INFRASTRUCTURE RECOMMENDATIONS

Internet Provider: A business-grade fiber plan (e.g., PLDT Business Fibr or Converge Business) with at least 100 Mbps symmetrical and an SLA guaranteeing uptime — critical since the IT/dev team needs reliable upload speed for GitHub pushes and client demos.

Server Specifications: Rack or tower server with at minimum a 6-core CPU, 32GB RAM, RAID 1/5 storage (2x1TB SSD minimum) — enough headroom to run Ubuntu Server, internal file sharing, and a small dev/test environment for 20 users.

Backup Strategy: Follow the 3-2-1 rule — 3 copies of data, on 2 different media (NAS + external drive), 1 stored offsite/cloud. Automate nightly incremental backups and weekly full backups.

Security Recommendations: Firewall with intrusion detection, network segmentation (separate VLANs for IT/servers vs. general staff), regular OS/software patching schedule, and restricted admin access.

Antivirus: Microsoft Defender for Business (already bundled with Windows 11 Pro/365) provides centralized endpoint protection without added licensing cost — suitable for a 20-person startup budget.

Password Policy: Minimum 12 characters, mix of upper/lower/numbers/symbols, mandatory change every 90 days, MFA enabled for email and admin accounts, no password reuse.

Expansion Plan: Design the network with extra switch ports and IP address headroom (e.g., /24 subnet) now, so the company can scale to 40–50 employees without re-cabling; consider migrating file storage to a hybrid cloud model as data grows.

PART 8: PERSONAL REFLECTION

In this project, I learned how to build a company's IT infrastructure from scratch, starting with nothing and figuring out exactly what hardware, software, and network equipment a 20-person company would actually need. I also learned what a System Administrator does in this kind of situation, especially how to size hardware requirements to a small company's headcount and how to design a network diagram topology that connects everything logically. Beyond the technical side, I learned that infrastructure planning also involves thinking about departments, budgets, and how the company might grow in the future.

The most challenging part of this project was planning the network topology for the company. It was difficult to figure out how each device, such as the router, firewall, switch, and access points, should logically connect to each department, while still making the diagram clear and easy to read. I had to revise the layout a few times before the connections made sense and matched how the departments would actually use the network.

Planning is important because, without it, the execution can lead to an unsecured network and a disorganized infrastructure for the company. For example, buying equipment without first mapping out the network could mean missing a firewall, running out of switch ports, or exposing the company to security risks that are expensive to fix later. Planning ahead also saves money, since it prevents a company from buying the wrong equipment or having to redo the network as it grows.

This project helped me develop critical thinking and problem-solving skills, especially in figuring out what a company actually needs before making any purchases. It gave me a small taste of what a real System Administrator does, planning infrastructure, justifying decisions, and thinking ahead about security and growth, which I will carry into future IT work. Overall, this project made me more confident in approaching real-world IT planning tasks, and it showed me how much thought goes into a company's infrastructure before a single computer is even purchased.